

CR TP1 Redis

3 Commands

3.1 Basic commands

1. Store and then retrieve a string "Aldo Maccione" associated to the key "actor:1" R: SET actor:1 "Aldo Maccione" et GET actor:1 et les retours sont respectivement OK et "Aldo Maccione".

2. Store in a list "movies:sw" the movies "The phantom menace" and "Attack of the Clones" R: LPUSH movies:sw "The phantom menace" et LPUSH movies:sw "Attack of the Clones" et les retours sont respectivement (integer) 1 et (integer) 2.

3. Return the length of the list "movies:sw" R: LLEN movies:sw et le retour est (integer) 2.

4. Return (and remove) the element from the head of the list "movies:sw" R: LPOP movies:sw et le retour est "The phantom menace".

3.2 Queues and stacks

1. Implement a queue (FIFO) using a list (for example with values 1, 2, 3, 4 and 5) using LPUSH. In particular how to design "enqueue" and "dequeue" R: LPUSH fifo 1 ou 2 ou 3 ou 4 ou 5 pour enqueue et RPOP fifo pour dequeue. Les retours sont (integer) 1, (integer) 2, (integer) 3, (integer) 4, (integer) 5 et "1" pour dequeue.

2. Implement a queue using RPUSH R: RPUSH fifo 1 ou 2 ou 3 ou 4 ou 5 pour enqueue et LPOP fifo pour dequeue. On fait juste l'inverse de LPUSH et RPOP.

3. Implement a stack (LIFO) using a list (for example with values 1, 2, 3, 4 and 5) using LPUSH. In particular how to design "push" and "pop" R: LPUSH stack 1 ou 2 ou 3 ou 4 ou 5 pour push et LPOP stack pour pop. Les retours sont (integer) 1, (integer) 2, (integer) 3, (integer) 4, (integer) 5 et "5" pour pop.

4. Implement a stack using RPUSH R: RPUSH stack 1 ou 2 ou 3 ou 4 ou 5 pour push et RPOP stack pour pop. On fait juste l'inverse de LPUSH et RPOP.

4 Data caching

4.3 Validation

4.3.1 Full result caching / Query caching Let the query `SELECT * FROM MOVIE WHERE RUNTIME=143` associated to the following records:

tt0325980,Pirates of the Caribbean: The Curse of the Black Pearl,2003-07-09,143,140000000,655011224,<<https://image.tmdb.org/t/p/original/z8onk7LV9Mmw6zKz4hT6pzzvmv1.jpg>>,"",
tt1477834,Aquaman,2018-07-06,143,160000000,1148461807,<<https://image.tmdb.org/t/p/original/xLPffWMhMj1l5.jpg>>,"",

1. Store the previous query as a string in the CSV format

```
SET "SELECT * FROM MOVIE WHERE RUNTIME=143" "tt0325980,Pirates of the Caribbean: The Curse of the Black Pearl,2003-07-09,143,140000000,655011224,<https://image.tmdb.org/t/p/original/z8onk7LV9Mmw6zKz4hT6pzzvmv1.jpg>","",
```

R: OK

2. Store the previous query as a string in the JSON format

```
SET "SELECT * FROM MOVIE WHERE RUNTIME=143" '[{"idmdb": "tt0325980", "title": "Pirates of the Caribbean: The Curse of the Black Pearl", "release": "2003-07-09", "runtime": 143, "budget": 140000000, "revenue": 655011224, "poster": "<https://image.tmdb.org/t/p/original/z8onk7LV9Mmw6zKz4hT6pzzvmv1.jpg>", "overview": "", "rating": "", "last_update": ""}, {"idmdb": "tt1477834", "title": "Aquaman", "release": "2018-07-06", "runtime": 143, "budget": 160000000, "revenue": 1148461807, "poster": "<https://image.tmdb.org/t/p/original/xLPffWMhMj1l5.jpg>", "overview": "", "rating": "", "last_update": ""}]'
```

4.3.2 Individual result caching / Tuple caching Let the 83 minute movie 'Snow White and the Seven Dwarfs', released on 1937-12-21, associated with the id tt0029583. Its budget was 1488423\$, its revenue is 184925486 \$ and its rating is 7.6. Its cover can be seen at <https://image.tmdb.org/t/p/original/1P9eGGlT7eV7kAAvvSh9jfygr1C.jpg>.

1. Store 'Snow White and the Seven Dwarfs' as a CSV string

SET "tt0029583" 'tt0029583,Snow White and the Seven Dwarfs,1937-12-21,83,1488423,184925486,<

2. Retrieve the size of this CSV like entry (memory size) R: la commande est MEMORY USAGE "tt0029583" et le résultat est (integer) 176

3. Store 'Snow White and the Seven Dwarfs' as a hash

HSET tt0029583 idmb tt0029583 title "Snow White and the Seven Dwarfs" release "1937-12-21" r

4. Retrieve the size of this hash entry R: la commande est MEMORY USAGE tt0029583 et le résultat est (integer) 534